
Detecting Patronizing & Condescending Language (PCL) with RoBERTa-large

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Code: <https://github.com/teobenarous/pcl-detection>

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1 Task Definition and Literature Review

1.1 Task Definition

Progress in Natural Language Processing (NLP) is often driven by shared tasks that benchmark model performance against specific linguistic challenges. The focus of this project is the *SemEval 2022 Task 4 (Subtask 1): Detecting Patronizing and Condescending Language (PCL)* challenge. The objective is a binary classification problem: given a sequence of text extracted from news articles, the model must predict whether the text contains PCL or not.

1.2 Literature Review

To contextualize this task, we review the foundational paper, « *Don't Patronize Me! An Annotated Dataset with Patronizing and Condescending Language towards Vulnerable Communities* » [1], which introduced the dataset used in this challenge.

Q1. Primary Contributions

The primary contribution of this work is the introduction of the « *Don't Patronize Me!* » dataset, the first annotated corpus explicitly designed to identify and categorize PCL directed at vulnerable communities. The authors formalize this novel challenge into two NLP tasks: predicting the presence of PCL via binary classification, and categorizing the specific type of PCL based on a newly proposed taxonomy (e.g., « The savior, » « The expert ») via multi-label classification. Additionally, they establish benchmark performances using various models, demonstrating that pre-trained language models like RoBERTa and BERT achieve the strongest baseline results.

Q2. Technical Strengths

A major technical strength is the paper's originality in addressing a subtle, often unconscious form of harm that contrasts sharply with well-studied, explicit hate speech. The dataset construction is robust and geographically diverse, extracting 10,637 paragraphs from the News on Web corpus across 20 English-speaking countries. Furthermore, the authors adhere to high standards of reproducibility and evaluation by detailing specific hyperparameters, providing a public GitHub repository, and appropriately using precision, recall, and F1 scores to account for the severe class imbalance (only 995 positive instances).

Q3. Key Weaknesses

Despite its strengths, the dataset's reliance on a rigid set of predefined keywords (e.g., « homeless, » « refugee ») introduces selection bias, inherently restricting the corpus to overt mentions and potentially missing implicit instances of PCL. The subjectivity of the task is also a limitation, evidenced by an inter-annotator agreement of 41% across three labels—which the authors characterize as a « moderate agreement » rather than a low one—suggesting that the « gold » labels inherently contain some ambiguity.

Synthesis

This paper successfully introduces a significant and nuanced NLP challenge, providing an annotated dataset that establishes a strong foundation for future research in subtle toxicity detection. While the keyword-driven methodology inherently introduces some selection bias and the extreme subjectivity of the task caps annotator agreement, the thorough baseline benchmarking and excellent reproducibility make this a high-quality contribution to the field.

2 Exploratory Data Analysis

To build a robust classifier and understand the linguistic properties, class balance, and potential noise within the dataset, a comprehensive Exploratory Data Analysis (EDA) was conducted. To ensure clarity, this section begins with a basic statistical profile, followed by two EDA techniques that informed our decision to prepend contextual metadata to the text sequences.

2.1 Basic Statistical Profiling & Data Hygiene

Before analyzing semantics, the structural integrity of the dataset was profiled:

- **Class Imbalance:** The negative class (no-PCL) contains 7,581 samples, while the positive class (PCL) contains only 794 samples—an imbalance ratio of approximately 9.5:1. (These figures reflect the training partition.)
- **Sequence Length:** The text length distributions are nearly identical across both classes (median 42 words, 75th percentile 60 words). Because RoBERTa utilizes a sub-word byte-pair encoding tokenizer, raw word counts naturally expand during tokenization. To safely accommodate this sub-word expansion, capture the longer tail of the sequence distribution, and provide capacity for the injected contextual metadata, a maximum length of 256 tokens was established.
- **Data Normalization:** Raw sequence strings exhibited various web-scraping artifacts that natively disrupt contextual embeddings. We implemented a text normalization pipeline to standardize the input space. This included resolving encoding errors, stripping residual HTML, and masking highly variable entities (URLs and social media handles) with generic [URL] and [USER] tokens to reduce embedding sparsity. Crucially, because BPE tokenizers are highly sensitive to arbitrary spacing, we performed rigorous de-tokenization—repairing fragmented English clitics (e.g., « ca n't » to « can't ») and standardizing anomalous typographical artifacts (e.g., isolated dashes and corrupted quotation marks). Finally, anomalous texts shorter than five words were discarded to eliminate uninformative caption noise.

2.2 EDA Technique 1: N-Gram and Lexical Profiling

Visual Evidence

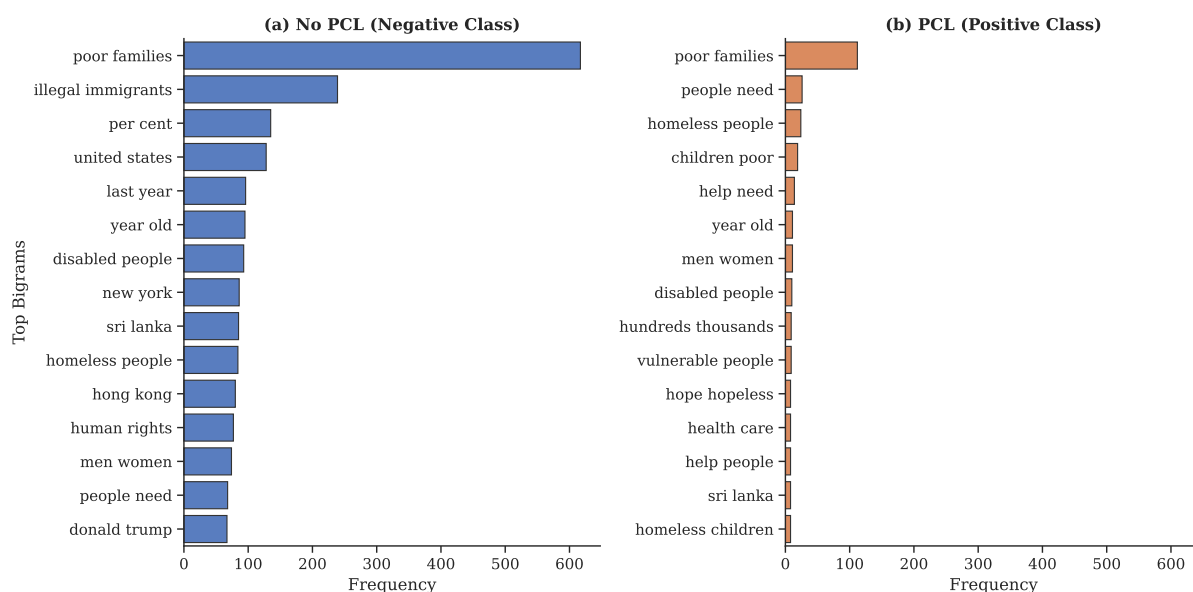


Figure 1: Top 15 most frequent bigrams for the no-PCL (left) and PCL (right) classes.

Analysis

Extracting the most frequent bigrams highlights lexical overlaps and semantic divergences. Common demographic identifiers are highly prevalent regardless of the class; for example, « poor families, » « homeless people, » and « disabled people » appear prominently in *both* distributions. The divergence between the classes manifests in their surrounding contextual framing. The no-PCL texts frequently utilize standard journalistic boilerplate (e.g., « per cent, » « last year ») and focus on broader political subjects (e.g., « illegal immigrants, » « united states »). Conversely, the PCL texts rely heavily on subjective editorializing, vulnerability, and savior-centric framing (e.g., « help need, » « vulnerable people »).

Impact Statement

The heavy overlap of target demographic terms across both classes suggests that the mere presence of a vulnerability keyword does not guarantee a patronizing intent; the classification is highly dependent on the surrounding syntactic context. This insight motivated the contextual metadata injection strategy, where the tabular metadata (the targeted [KEYWORD]) is prepended to the text sequence. This structural modification provides the model with immediate contextual grounding, allowing the attention mechanism to condition its representations on the target demographic before parsing the sequence semantics.

2.3 EDA Technique 2: Named Entity Recognition (NER) & Geographic Bias

Visual Evidence

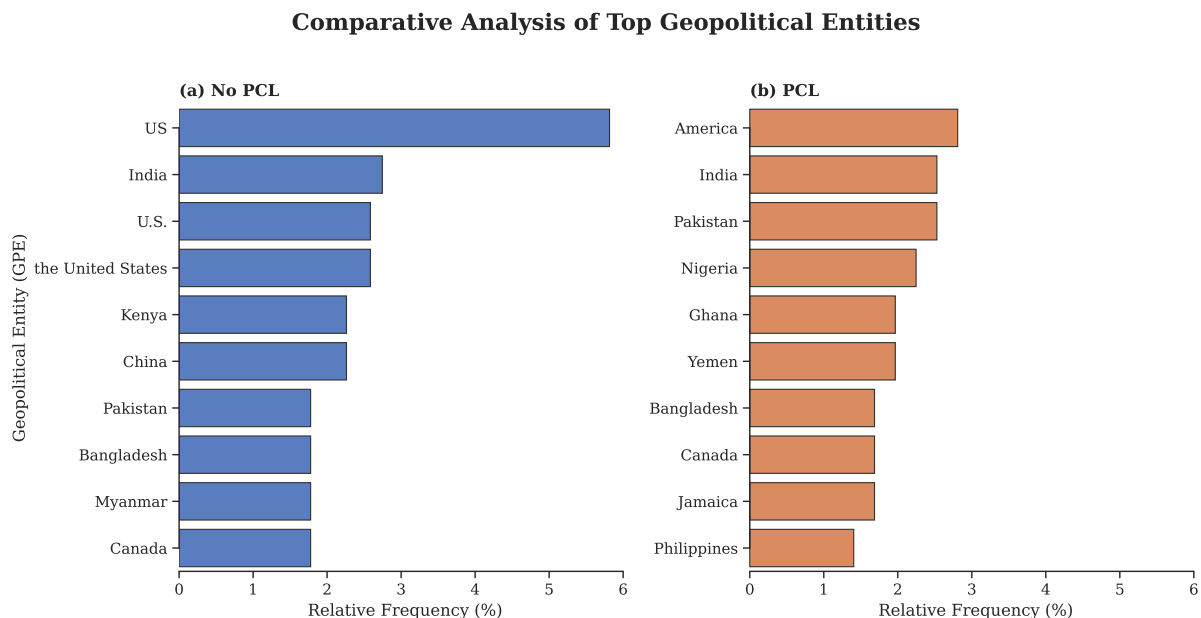


Figure 2: Comparative analysis of top geopolitical entities extracted via NER.

Analysis

Applying NER to extract geopolitical entities reveals a distinct shift in geographic distribution between the two classes. Both classes share significant commonalities, frequently referencing North America and India. However, the PCL distribution flattens significantly and introduces a pronounced « long tail » of developing nations (e.g., Nigeria, Ghana, Yemen, Jamaica, Philippines). These specific Global South nations are highly prevalent in patronizing contexts but are absent from the top ten entities in standard, non-patronizing reporting.

Impact Statement

This variance empirically supports the thesis that patronizing journalistic framing is often correlated with the specific region being discussed. Recognizing that geographic bias acts as a latent feature of condescension, this finding justified the expansion of the context enrichment strategy to include prepending the [COUNTRY] code into the model's input sequence.

3 Experimental Baseline and Proposed Approach

3.1 Data Partitioning Strategy

While the shared task organizers provided standard training and development splits, the official test set labels remained hidden for the final evaluation. Relying solely on the provided development set for both model selection and final local evaluation introduces data leakage and yields over-optimistic performance estimates.

To establish a methodologically sound evaluation framework, the official training and development sets were merged. An 80/10/10 stratified split was then performed. Consequently, our training set overlaps with the official development split; however, this was a deliberate methodological choice to maximize the volume of training data, rather than depleting the original training split to form an internal validation set. This partition yielded:

- **Training Set (80%):** Utilized strictly for model training.
- **Internal Validation Set (10%):** Utilized as the in-training proxy for early stopping, model selection, and calibrating the decision threshold.
- **Internal Test Set (10%):** Held out entirely during the training and validation phases to serve as an unbiased, unseen proxy for the unlabeled external test set, utilized exclusively for the final local evaluation.

Stratification was applied across all splits to ensure that the 9.5:1 class imbalance identified during the EDA was perfectly preserved in every data partition.

3.2 Reference and Experimental Baselines

The standard reference point for this task is the RoBERTA-BASE model provided by the SemEval organizers, which achieved an F1 score of 0.48 on the official development set and 0.49 on the official test set.

Due to the inherent difficulty and subtle semantic nuances of identifying patronizing language, a stronger experimental baseline was established utilizing a scaled-up architecture (RoBERTA-LARGE, 355M parameters). This baseline serves as the control for all subsequent ablations and includes foundational upgrades derived directly from the Exploratory Data Analysis:

- **Data Hygiene:** Application of the text cleaning pipeline detailed in Section 2.
- **Contextual Metadata Injection:** Prepending tabular demographic data ([KEYWORD] and [COUNTRY]) into the text sequences.
- **Prior-Probability Bias Initialization:** Calibrating the classification head by initializing the final linear layer’s bias to $b = -\log((1 - \pi)/\pi) \approx -2.25$, where π is the positive class prior (≈ 0.095). This intervention bypasses the base-rate « burn-in » phase, allowing the network to immediately begin learning linguistic features rather than spending early epochs iteratively correcting its default uniform prior assumption.

What varies across the ablation runs are exclusively two targeted interventions: layer-wise learning rate decay and an alpha-balanced focal loss.

3.3 Layer-wise Learning Rate Decay (LLRD)

Fine-tuning a massive 24-layer transformer on a relatively small dataset risks « catastrophic forgetting, » where robust, pre-trained linguistic structures in the early layers are destructively overwritten by task-specific gradients. To mitigate this, we implemented LLRD. Rather than applying a uniform learning rate, LLRD applies a multiplicative decay factor ($\lambda = 0.95$) sequentially layer-by-layer, starting from the network output down to the input.

Consequently, the top layers (those closest to the classification head) adapt aggressively to the subtle PCL task at the full base learning rate. Conversely, the bottom layers (closest to the input embeddings)—which process fundamental English syntax—receive a progressively decayed learning rate. Across 24 transformer layers, the foundational embedding layer retains only about 29% (0.95^{24}) of the base learning rate. This architectural intervention ensures smooth gradient flow, aggressively tuning task-specific semantics while preserving structural English syntax.

3.4 Alpha-Balanced Focal Loss

The standard 9.5:1 class imbalance of the PCL dataset fundamentally destabilizes traditional classification approaches. Standard interventions typically involve either data-level or loss-level adjustments. However, common implementations within these categories possess severe flaws in this context:

- (a) **Weighted Random Sampling (Oversampling):** Resolving the imbalance via oversampling requires the network to process the exact same minority examples roughly 9 to 10 times per epoch. Feeding identical, duplicated sequences to a highly parameterized architecture like RoBERTA-LARGE would most likely cause overfitting rather than the learning of generalizable linguistic features.
- (b) **Data Augmentation via Backtranslation:** While synthetic text generation is a standard data-level mitigation for imbalance, backtranslation neutralizes the highly subtle phrasing and implicit condescension inherent to PCL. This introduces severe semantic shift and noisy labels into the minority class, deteriorating data quality.
- (c) **Weighted Cross-Entropy (WCE):** WCE attempts to solve the imbalance by applying a static, global penalty based on the class. However, it treats all examples within a class equally. In a 9.5:1 imbalance, the thousands of «easy» majority examples (which the model already predicts correctly with high confidence) still cumulatively dominate the gradient, blinding the model to subtle boundary cases.

To address these shortcomings simultaneously, standard cross-entropy was replaced with an α -balanced Focal Loss (FL). FL reshapes the loss space by dynamically down-weighting the penalty assigned to easily classified examples, forcing the model to focus entirely on hard, borderline instances:

$$\text{FL}(p_n) = -\alpha_n(1 - p_n)^\gamma \log(p_n),$$

where p_n is the model’s estimated probability for the ground-truth class. The factor α_n is the instance-specific weight, defined as the hyperparameter α for the positive class and $(1 - \alpha)$ for the negative class. The modulating factor $(1 - p_n)^\gamma$ zeros out the loss as the model’s confidence increases ($p_n \rightarrow 1$).

While canonical implementations utilize a focusing parameter of $\gamma = 2$ for extreme imbalances (e.g., 1000:1), this dataset relies heavily on learning «hard negatives»—empathetic but non-condescending sentences concerning vulnerable communities. At $\gamma = 2$, the loss signal from these crucial semantic boundary cases is overly suppressed. By reducing the focusing parameter to $\gamma = 1$, a meaningful learning signal is preserved for these hard negatives while still mitigating the overwhelming influence of the trivially easy majority examples.

Given $\gamma = 1$, the class weight α must be carefully calibrated. While an exhaustive hyperparameter sweep across the 355M parameter architecture would be ideal, hardware constraints rendered a massive grid search computationally prohibitive. Instead, α was derived by targeting an approximately balanced training gradient (i.e., a 50% positive gradient fraction). Based on the dataset’s prior and the expected loss of borderline cases, the optimal weight was calculated as $\alpha \approx 0.745$. Consequently, the focal loss was configured with $\gamma = 1$ and $\alpha = 0.75$, slightly encouraging positive samples.

3.5 Explored but Excluded Alternatives

In designing this pipeline, additional alternative methodologies were explored but ultimately discarded:

- **Task-Adaptive Pretraining (TAPT):** Continuing Masked Language Modeling on the target corpus before fine-tuning is theoretically sound for domain adaptation. However, the PCL dataset is remarkably small ($\sim 10,000$ sequences). Performing MLM on such a limited corpus using a 355M parameter model offers negligible semantic shifts while introducing a severe risk of overfitting before the classification head is even initialized.
- **Supervised Contrastive Learning (SupCon):** Discarded due to its documented failure on binary imbalanced distributions. Mildenerger et al. [2] demonstrate empirically that SupCon’s performance degrades monotonically with increasing class imbalance, falling below standard cross-entropy at moderate imbalance levels. They attribute this to gradient saturation: the output gradient for each sample is theoretically upper-bounded by the inverse of the number of positives in the batch, causing minority-class embeddings to receive vanishingly small updates and the representation space to collapse toward the majority class cluster.

4 Implementation Details

4.1 Experimental Setup and Hyperparameters

To ensure reproducibility, the implementation leveraged PyTorch and the HuggingFace transformers library. Early stopping and model selection across epochs were governed by monitoring the area under the precision-recall curve (AUPRC) on the internal validation set. The complete configuration utilized for the proposed approach is detailed in Table 1.

Hyperparameter	Value
Base Architecture	ROBERTA-LARGE
Maximum Sequence Length	256 tokens
Per-Device Batch Size	8
Gradient Accumulation Steps	4
Hardware Precision	BF16
Optimizer	AdamW
Base Learning Rate	1×10^{-5}
Weight Decay	0.01
Warmup Ratio	0.1
LLRD Decay Factor (λ)	0.95
Focal Loss Gamma (γ)	1.0
Focal Loss Alpha (α)	0.75
Bias Initialization (b)	-2.25
Maximum Epochs	10
Early Stopping Patience	3

Table 1: Configuration for the proposed approach.

4.2 Ensemble Strategy

To mitigate initialization variance and prevent overfitting, a 5-seed ensemble of the best-performing architecture was trained. For aggregation, soft voting—averaging the raw predicted probabilities across all five models prior to thresholding—was adopted in preference to hard voting. Hard voting requires each constituent model to discretize its probability estimates into binary predictions before aggregation, irreversibly discarding the calibrated confidence signal. Soft voting preserves this information, allowing the ensemble to make a more informed collective decision and producing a better-calibrated predictive distribution.

4.3 Reproducibility

The repository can be accessed at: <https://github.com/teobenarous/pcl-detection>.

5 Evaluation and Error Analysis

Evaluating an imbalanced classification model solely on F1 score obscures the underlying calibration properties of the model. While the positive-class F1 serves as the primary benchmark metric for this task, a practically viable classifier must demonstrate robust confidence calibration—measured jointly by precision, recall, the decision boundary (τ), and the area under the precision-recall curve (AUPRC). Under considerable class imbalance (e.g., 9.5:1), AUPRC constitutes a superior evaluation metric to the area under the receiver operating characteristic (AUROC), as AUROC incorporates true negatives, which artificially inflates performance estimates when the majority class dominates.

5.1 Ablation Analysis & Model Selection

Model / Experiment	Split	Threshold	AUPRC	Acc.	C0 F1	C1 Prec.	C1 Rec.	C1 F1
Baseline	Internal Val.	0.1037	0.6931	0.9263	0.9589	0.6000	0.6900	0.6419
	Internal Test	0.1037	0.6147	0.9187	0.9546	0.5583	0.6768	0.6119
LLRD Only	Internal Val.	0.0197	0.6791	0.9215	0.9560	0.5714	0.7200	0.6372
	Internal Test	0.0197	0.6045	0.9167	0.9533	0.5469	0.7071	0.6167
Focal Loss Only	Internal Val.	0.4797	0.6960	0.9445	0.9699	0.8281	0.5300	0.6463
	Internal Test	0.4797	0.6504	0.9321	0.9632	0.7258	0.4545	0.5590
LLRD + Focal Loss	Internal Val.	0.5898	0.6742	0.9301	0.9614	0.6364	0.6300	0.6332
	Internal Test	0.5898	0.6221	0.9330	0.9630	0.6465	0.6465	0.6465
Ensemble (Soft Voting)	Internal Val.	0.4900	0.7190	0.9416	0.9677	0.6931	0.7000	0.6965
	Internal Test	0.4900	0.6704	0.9301	0.9612	0.6182	0.6869	0.6507

Table 2: Ablation and ensemble evaluation metrics across internal validation and test splits. (C0 = no-PCL, C1 = PCL).

The Brittle Baseline

The baseline’s test F1 (0.6119) rests on a miscalibrated decision boundary ($\tau = 0.1037$), indicating that the model’s output probabilities are compressed toward low values across the board. The resulting test precision-recall profile (0.5583 / 0.6768) reflects an indiscriminate classification policy: the model achieves coverage by casting a wide net rather than through genuine discriminative confidence. The test AUPRC (0.6147) serves as the reference for the model’s inherent separability prior to any thresholding.

Analyzing the Enhancements: Complementary Failure Modes

Focal loss shifts the probability distribution toward a well-calibrated threshold, substantially improving precision and AUPRC at the cost of recall, with a severe generalization gap indicating overfitting. LLRD produces the opposite effect: recall improves while precision degrades slightly, calibration worsens, and AUPRC marginally decreases relative to the baseline.

The Synergy: LLRD as a Structural Regularizer for Focal Loss

The two enhancements are complementary by construction: focal loss corrects calibration and improves precision, while LLRD recovers recall and mitigates overfitting. Combined, they yield a well-calibrated model with a balanced precision-recall profile, the best single-model test F1, and an AUPRC that exceeds the baseline and LLRD in isolation, though it falls below the focal-only configuration.

The Soft Voting Ensemble

The soft voting ensemble extends the single-model approach by averaging raw probabilities across five initialization seeds, reducing variance and producing a more stable aggregated distribution. This yields modest but consistent gains: test F1 improves modestly and test AUPRC significantly, with a well-centered decision boundary.

5.2 Local Evaluation: Custom Metrics

To visually understand how the soft voting ensemble outperformed the baseline, their respective confusion matrices and precision-recall (PR) curves were evaluated on the internal test set (Figure 3).

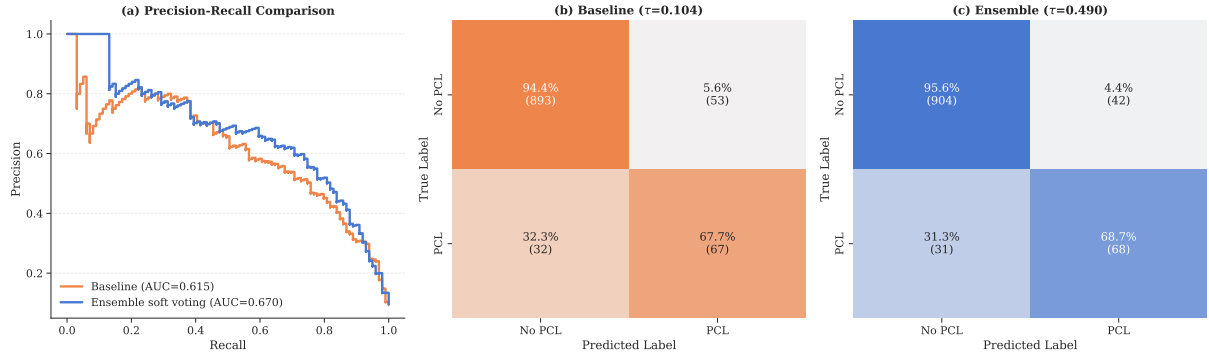


Figure 3: Precision-recall curves and confusion matrices evaluated on the internal test split.

The confusion matrices explicitly detail how the ensemble is superior to the baseline. The baseline identified 67 true positives (recall of 67.7%) but at the cost of 53 false positives, reflecting its uncalibrated decision boundary. Conversely, the soft voting ensemble utilized a calibrated, nearly centered decision boundary; and despite requiring higher certainty to trigger a positive classification, the ensemble successfully filtered out over 20% of the baseline’s false positive errors (reducing them from 53 to 42) while simultaneously *increasing* its true positive detections (identifying 68 instances).

Furthermore, the ensemble’s PR curve sits visibly higher than the baseline’s across nearly the entire operational recall range, demonstrating that the synergistic approach improved the ability to separate the two classes.

5.3 Error Analysis: Linguistic Failure Modes

No.	Para.	Gold	Prob. (Baseline)	Prob. (Ensemble)
1	[KEYWORD] <i>migrant</i> [COUNTRY] <i>bd</i> [SEP] <i>At Davos, the World Economic Forum dedicated this year to cultivate “responsive” and “responsible” leadership in a world largely lacking both. This mission obviously refers to domestic-national leaders, who must now serve as the bastion of bringing a torn and submerging planet from disaster, whether caused by over-adrenalised presidents and prime ministers, or environmental decay, damage, and neglect, or something else. Clearly, many would be needed, for example, to heal the Syrian scars, home the burgeoning refugee-migrant influxes, and heed the U.N. SDG (Sustainable Development Goals) mandate to uplift the deprived, minorities, unschooled children, other suppressed women, and so forth.</i>	✓	0.086 (×)	0.900 (✓)
2	[KEYWORD] <i>homeless</i> [COUNTRY] <i>jm</i> [SEP] <i>Old BC: As usual, Old GB has it assy-versy. I think there should be more focus on preventing homelessness than curing it. Are there systems for actually going out, picking up the homeless, and placing them where they belong? Homeless people aren’t voluntarily walking into these shelters.</i>	✓	0.501 (✓)	0.401 (×)
3	[KEYWORD] <i>hopeless</i> [COUNTRY] <i>us</i> [SEP] <i>If this bill passes, our criminal justice system could be significantly altered for the better, and a path toward teshuvah created, rather than merely perpetuating a cycle of harsh punishments, recidivism and hopelessness. I call on people of faith, and all those who believe we must do better for the most vulnerable in our communities, to join me in urging the New Jersey Assembly to pass this vital legislation.</i>	×	0.558 (✓)	0.885 (✓)

Table 3: The table displays three representative paragraphs, their ground truth labels, and the raw predictive probabilities from the baseline ($\tau = 0.1037$) and ensemble ($\tau = 0.4900$) models (✓ = PCL, × = No PCL).

The Correction

This text is a manifestation of the « savior » taxonomy. The condescension lies in framing global leaders as « bastions » whose explicit mandate is to « uplift the deprived » and « heal the Syrian scars. » The baseline failed to flag this paragraph (8.6%), as the vulnerability keywords are buried within diplomatic jargon and lengthy clauses. The ensemble, by contrast, was overwhelmingly confident (90%), consistent with having learned to capture long-range dependency patterns beyond localized keyword proximity.

The Over-Correction

Example 2 highlights a limitation of a calibrated decision boundary. This text is patronizing in that it strips the vulnerable group of agency entirely (« picking up the homeless, and placing them where they belong »). The baseline flagged it easily (50.1%) due to the surface-level co-occurrence of « homeless » and « shelters. » The ensemble, however, under-predicted it (40.1%), falling short of its $\tau = 0.4900$ threshold. Because the text is framed as sociopolitical debate and lacks overt savior vocabulary, the ensemble lacked the certainty to trigger a positive classification. This illustrates a limitation: a higher decision boundary can suppress predictions on instances whose condescension is implicit rather than overt.

Shared Hallucinations

This example exposes a core limitation: the models successfully index on the lexicon of vulnerability but fundamentally struggle with pragmatic intent. This text represents standard political advocacy; the speaker is rallying support for criminal justice reform. However, both models hallucinated with high confidence. The phrase « do better for the most vulnerable in our communities » mirrors the savior taxonomy, yet in the context of political legislation, annotators deemed it objective advocacy rather than condescension. This demonstrates that heavily parameterized models still index heavily on lexical density of vulnerability and struggle to separate patronization from genuine political rhetoric.

6 Conclusion

This project successfully addressed the complex challenge of detecting Patronizing and Condescending Language (PCL). Guided by an exploratory data analysis, the pipeline included a text normalization and contextual metadata injection to equip the model to handle the geographic and demographic biases inherent to the dataset.

To rectify the shortcomings of naive classification approaches under a severe 9.5:1 class imbalance, the classification head was calibrated using prior-probability bias initialization. Furthermore, to bypass the catastrophic memorization risks of oversampling and the global gradient penalties of weighted cross-entropy, an α -balanced focal loss was employed. Integrating this loss function with layer-wise learning rate decay preserved pre-trained linguistic features while enhancing the model's generalization across the minority class.

Rigor was maintained throughout the training lifecycle via early stopping based on AUPRC and dynamic decision threshold tuning based on F1 validation scores. Finally, by evaluating a soft voting ensemble, initialization variance was successfully mitigated. This synergistic approach successfully restored calibration, maximized the area under the precision-recall curve and F1 score, and established a robust solution for subtle toxicity detection in Natural Language Processing.

References

- [1] Carla Pérez-Almendros, Luis Espinosa Anke, and Steven Schockaert. « Don't patronize me! an annotated dataset with patronizing and condescending language towards vulnerable communities ». In: *Proceedings of the 28th International Conference on Computational Linguistics*. 2020, pp. 5891–5902.
- [2] David Mildenerberger et al. « A tale of two classes: adapting supervised contrastive learning to binary imbalanced datasets ». In: *Proceedings of the Computer Vision and Pattern Recognition Conference*. 2025, pp. 10305–10314.

A Additional Figures

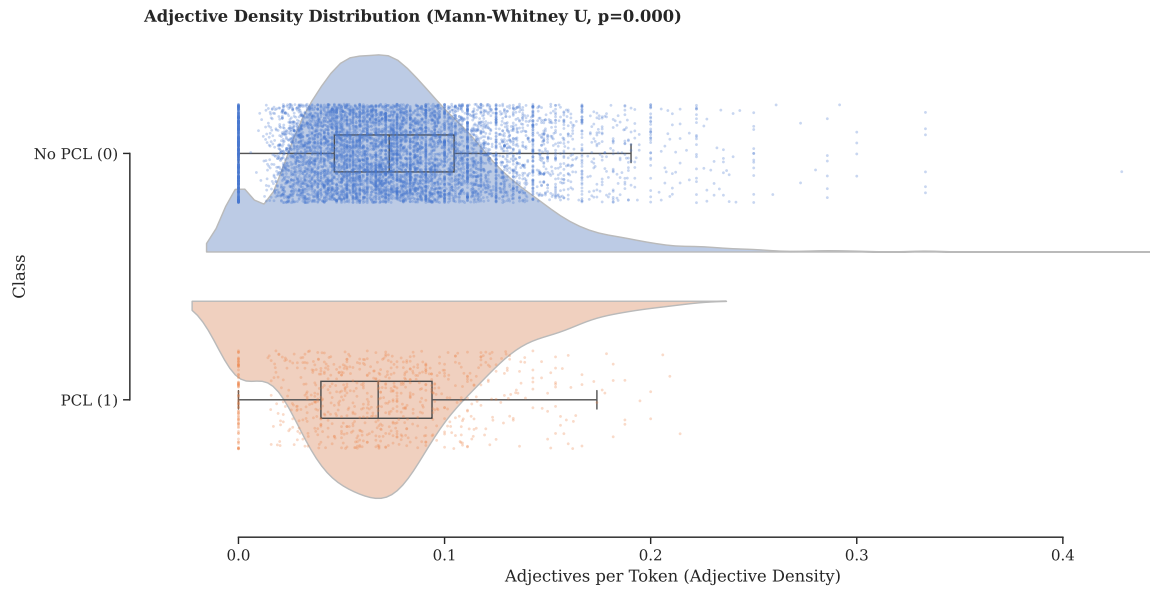


Figure 4: Raincloud plots illustrating the distribution of adjective density (adjectives per token) across the no-PCL and PCL classes. A Mann-Whitney U test indicates a statistically significant difference between the distributions ($p < 0.001$). However, the heavy visual overlap suggests that while condescending language may be marginally more adjective-dense, this feature alone is insufficient as a strong discriminatory signal for classification.

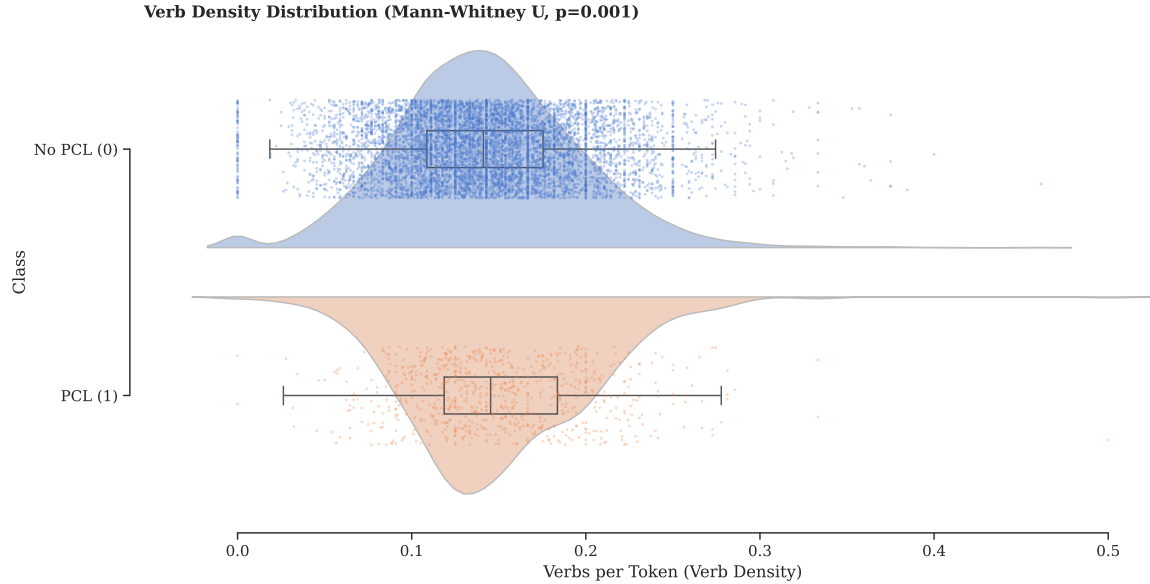


Figure 5: Raincloud plots comparing verb density across both classes. Similar to adjective density, a Mann-Whitney U test reveals a statistically significant difference ($p = 0.001$). Nevertheless, the substantial overlap in the distributions demonstrates that baseline verb frequency provides limited predictive power in isolation.

Semantic Separation in RoBERTa Embeddings



Figure 6: A 2D t-SNE projection of the raw, pre-trained RoBERTa-LARGE embeddings for the training set. Although the model groups the text into two distinct spatial clusters (likely based on structural or lexical features), the labels are intermixed within both. It highlights the difficulty of the task.



Figure 7: Word clouds visualizing the most frequent unigrams in the no-PCL and PCL classes. While common demographic terms overlap, the PCL class exhibits a higher concentration of savior-centric vocabulary (e.g., « help, » « need »).

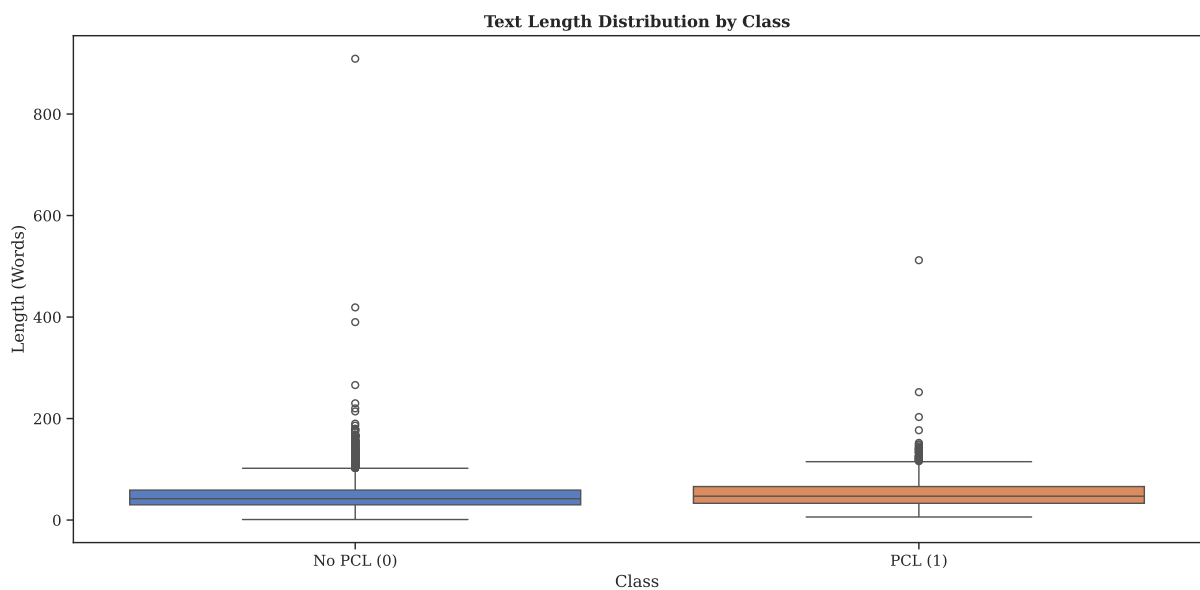


Figure 8: Boxplots displaying the text length distribution (in words) for each class. Both classes exhibit nearly identical length distributions with medians around 42 words and extensive long-tail outliers.